AUTONOMOUS WAREHOUSE ROBOT TASK MANAGEMENT SYSTEM

Algorithm

1. Start.

2. Initialize an empty queue named **pendingTasks** and an empty stack named **completedTasks**.

3. Display the main menu and accept the user's choice.

4. If choice = 1, read a task description and enqueue it if it is not empty.

5. If choice = 2, check the queue. If empty, show a message; otherwise dequeue the oldest task, simulate processing, and push it onto the completed-task stack.

6. If choice = 3, display all pending tasks by copying the queue and removing items from the copy.

7. If choice = 4, display completed-task history by copying the stack and popping items from the copy.

8. If choice = 5, check the stack. If empty, show a message; otherwise display and pop the latest history entry.

9. If choice = 6, display the exit message and terminate.

10. For any other choice, display an invalid-choice message.

11. Repeat the menu and selected operation until the user selects Exit.

12. Stop.